

Muhammad Talha Zaheer

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PROFESSIONAL SUMMARY

Computer scientist specializing in Agentic AI and secure software architecture. Experienced in building autonomous AI solutions with Next.js, Node.js, and the Vercel AI SDK—including tool calling, streaming generation, and production deployment. Combines full-stack development with a defensive security mindset shaped by hands-on cybersecurity training and instruction. Passionate about technical education, community leadership, and designing intelligent systems that are both innovative and secure.

TECHNICAL SKILLS

Languages	Python, JavaScript, TypeScript, Bash, SQL
Frameworks	Next.js, React, Node.js
AI & ML	Agentic AI, LLMs, Vercel AI SDK, Tool Calling, Pipecat AI, Deepgram, Cartesia, Daily PSTN, Prompt Optimization, LM Studio
DevOps & Deployment	Git/GitHub, GitHub Actions, Docker, CI/CD, Vercel, Railway, Digital Ocean, Supabase
Tools & IDEs	Postman, Cursor, VS Code, Google Antigravity, VirtualBox
Project Management	Monday.com, Linear

PROFESSIONAL EXPERIENCE

Core Edge Solutions	<i>May 2025 – May 2026</i>
<i>AI Developer</i>	<i>Lahore, Pakistan</i>

- Built production voice agents using Retell and Pipecat AI, integrating Deepgram (STT), OpenAI (LLM), and Cartesia (TTS) for cost-efficient conversational calling.
- Implemented tool calling for call transfer, call end, voicemail detection, and external API integration; optimized Pacifica Bot prompts after learning TMS domain workflows.
- Developed *Zuum* (FastAPI + React), an internal platform that outbound-calls dispatch drivers for shipment status updates.
- Designed GitHub Actions CI/CD pipelines to automate deployments on Railway.
- Built Puppeteer automations for Trinium TMS: adding container notes, updating empty container status, and processing acknowledged containers.
- Wrote SQL queries against Trinium TMS, exposed data through custom APIs, and compiled reports for container tracking.

Professional Freelancing Training Program	<i>Nov 2024 – Apr 2025</i>
<i>Cybersecurity Intern</i>	<i>Lahore, Pakistan</i>

- Conducted web application penetration testing for SQL injection, file inclusion, and unrestricted file upload vulnerabilities.
- Performed network reconnaissance with Nmap and analyzed HTTP/traffic flows using Burp Suite and Wireshark.
- Researched and lab-tested exploits including EternalBlue (MS17-010), macro-based attacks, and GHDB-driven reconnaissance; tracked emerging CVEs for early evaluation.
- Configured WAF rules and pfSense firewalls to protect web applications; practiced digital forensics on hard drives and disk images.

RELEVANT PROJECTS

Voice Agent	<i>AI Agent</i>
Built production voice agents with Pipecat AI, Deepgram STT, OpenAI, and Cartesia TTS; implemented	

tool calling for transfers, voicemail detection, and API integrations.

POS System

Professional Work

Electron.js point-of-sale system for restaurants with order management, multi-printer support, and production deployment.

Waiter App

Professional Work

Staff-facing application for taking and syncing restaurant orders with the POS backend across service workflows.

CTF Platform

Web Application

Web platform for hosting and managing Capture The Flag cybersecurity competitions with practical challenge workflows.

File Sync System

C++ Application

C++ file synchronization utility for reliable local file management and structured data transfer.

Portfolio: talhazaheer.dev/#projects

EDUCATION

University of Management and Technology (UMT)

Nov 2019 – Aug 2025

Bachelor of Science, Computer Science

Lahore, Pakistan

Professional Freelancing Training Program (NIAIS)

Jun 2024 – May 2025

Diploma, Cyber Security

Lahore, Pakistan

NAVTTTC

Apr 2025 – Jul 2025

Certification, Digital Forensic & Cyber Security

Pakistan

ADDITIONAL EXPERIENCE

Hunarmand Punjab

Sep 2025 – Oct 2025

Course Instructor

Punjab, Pakistan

Delivered structured technical training as part of a government skills development initiative.